

DIFFUSION

INTRODUCTION

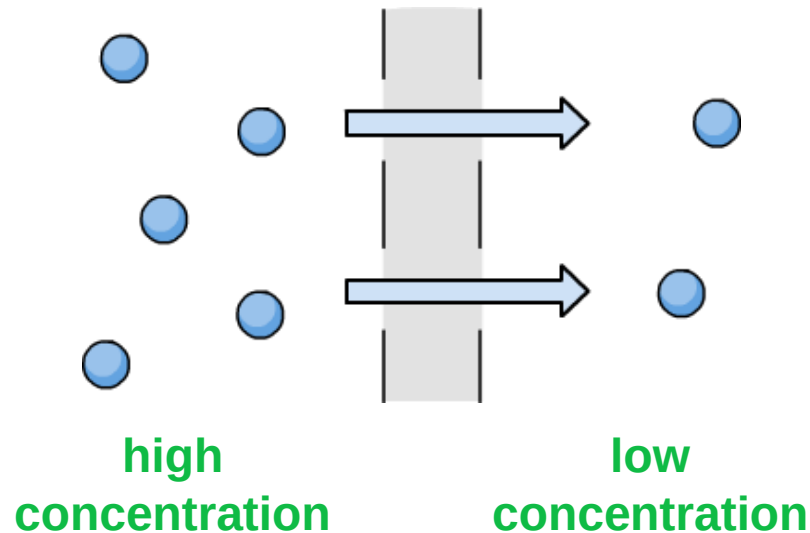
DEFINITION:

Diffusion:

- “The movement of particles from an area of **high concentration** to an area of **low concentration**, resulting in the **uniform distribution** of the substance.”
- During diffusion molecules move from an area of high concentration to an area of low concentration.
- They are said to move down a concentration gradient.
- The material that undergoes the transport is known as **diffusant or permeant or penetrant**.

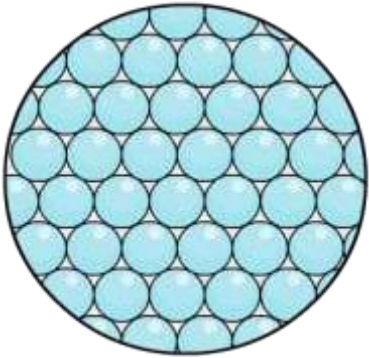
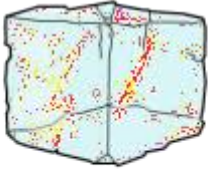
INTRODUCTION

- Diffusion is a **passive** process which means that **no energy** is needed.
- Molecules diffuse until they are evenly spaced apart and **equilibrium** is reached.

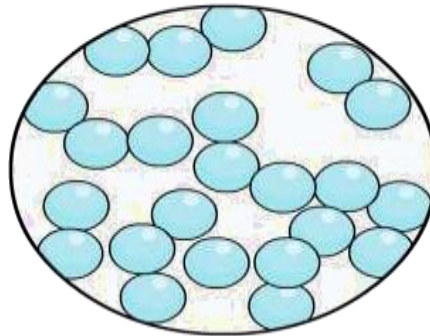


INTRODUCTION

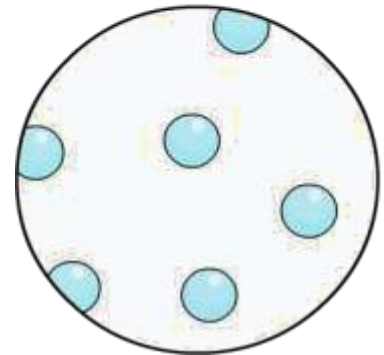
□ In which states are molecules able to diffuse?



solid (e.g. ice)



liquid (e.g. water)

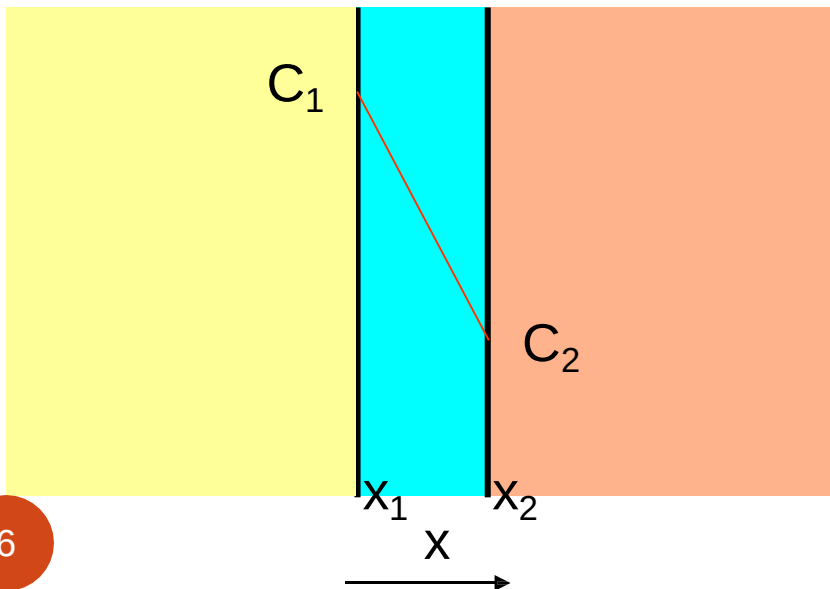


gas (e.g. steam)

Molecules in **liquids and gases** are constantly moving and bumping into each other. This means that they tend to spread out. By contrast, **solids** cannot diffuse.

STEADY STATE DIFFUSION

- Diffusion is direct result of **Brownian** motion.
- Molecule diffuse spontaneously from region of higher concentration to region of lower concentration until diffusion equilibrium is established.
- Rate of diffusion **independent** of time.
- **Flux** proportional to concentration gradient = $\frac{dC}{dx}$



Fick's first law of diffusion

$$J = -D \frac{dC}{dx}$$

D = diffusion coefficient

STEADY- STATE DIFFUSION

- The flux **J** is expressed as either in number of atom per unit time.
- Steady state diffusion means that **J** does not depend upon time.
- In this case fick's 1st law hold that flux along direction "x" is-

$$J = -D \frac{dC}{dx}$$

Where, J is equal to rate of mass transfer across unit surface area of barrier.

STEADY-STATE DIFFUSION

SINK CONDITION:

- State in which concentration in receptor compartment is **maintained** at lower level compared to its concentration in donor compartment.
- During diffusion donor compartment act as source & receptor compartment act as sink.
- Useful for maintaining concentration gradient nearly constant.

STEADY-STATE DIFFUSION

Fick's 1st law:

- **Mass get transported** from 1 compartment to another over period of time is Flux.
- Flux (J) = Rate of mass transfer across unit surface area of barrier.

$$J = \frac{1}{S} \left(\frac{dm}{dt} \right) \text{ ————— } 1)$$

Where,

dm = change in mass of material.

S = barrier surface area.

dt = change in time.

STEADY-STATE DIFFUSION

Acc.to Fick's law, flux is directly proportional to conc.gradient.

$$J = -D \frac{dC}{dx} \quad \text{—————} \quad 2)$$

D=diffusion coefficient of penetrant.

dx=change in distance.

Combine equation 1) & 2) gives,

$$dm/dt = -DS (dc/dx)$$

* This equation represent rate of mass transfer as per fick's law.

MECHANISM

Mechanisms:

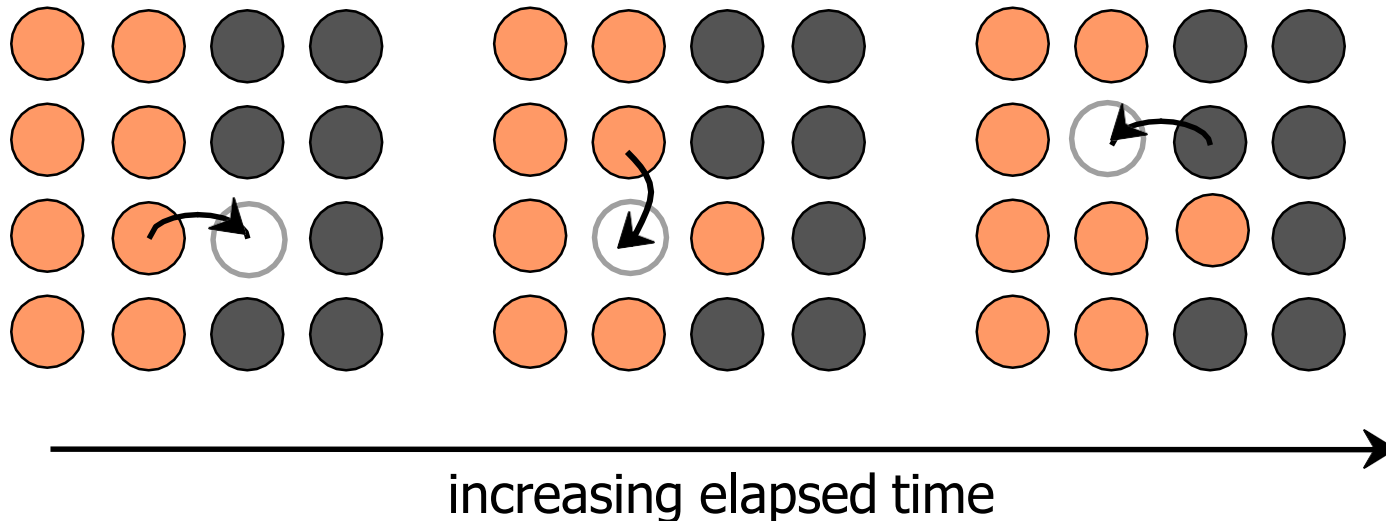
- **Gases & Liquids** – random (Brownian) motion.
- **Solids** – vacancy diffusion or interstitial diffusion.

I. Vacancy diffusion:

- Only **adjacent** atoms can **move** into a vacancy.
- Vacancy moves in **opposite direction of atomic motion**.
- Rate depends on concentration of vacancies.

MECHANISM

- atoms exchange with vacancies .
- **rate depends on:**
 - **-- number of vacancies**
 - **-- activation energy to exchange.**

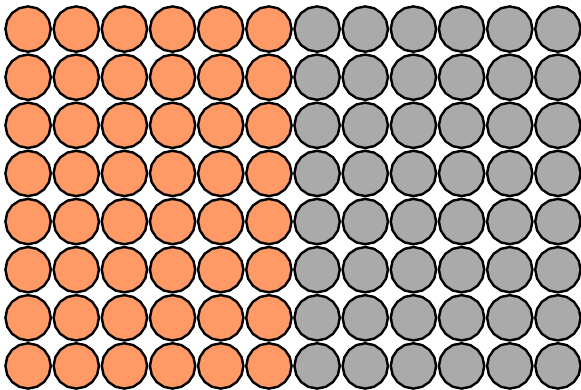


MECHANISM

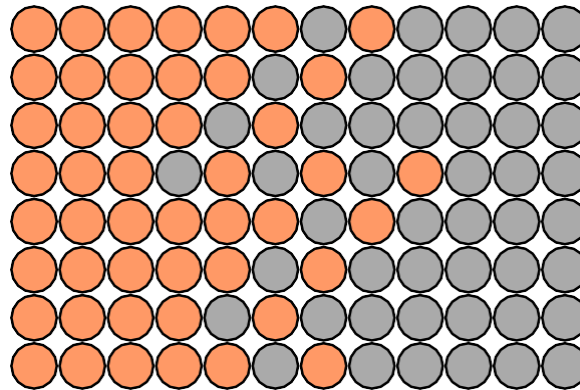
Inter diffusion: Atoms tend to **migrate** from regions of high conc. to regions of low concentration.

- – smaller atoms can **diffuse** between atoms.

Initially



After some time



- Atom can **move** into any adjacent empty interstitial position (usually smaller atoms).
- **Rate depends on concentration of interstitial atoms.**
- (Usually **faster** than vacancy diffusion).

Factors that Influence Diffusion

I. Diffusing Species:

- Magnitude of diffusion coefficient, D – indicates the rate at which atoms diffuse.
- Both diffusing species and host material influence the coefficient:
 - ✓ Relative sizes of atoms.
 - ✓ Openness of lattice.
 - ✓ Ionic charges.

Factors that Influence Diffusion

II. Temperature:

- Very strong effect on the diffusion coefficient:
- **Diffusion coefficient increases with increasing T.**

$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

D = diffusion coefficient

D_o = pre-exponential

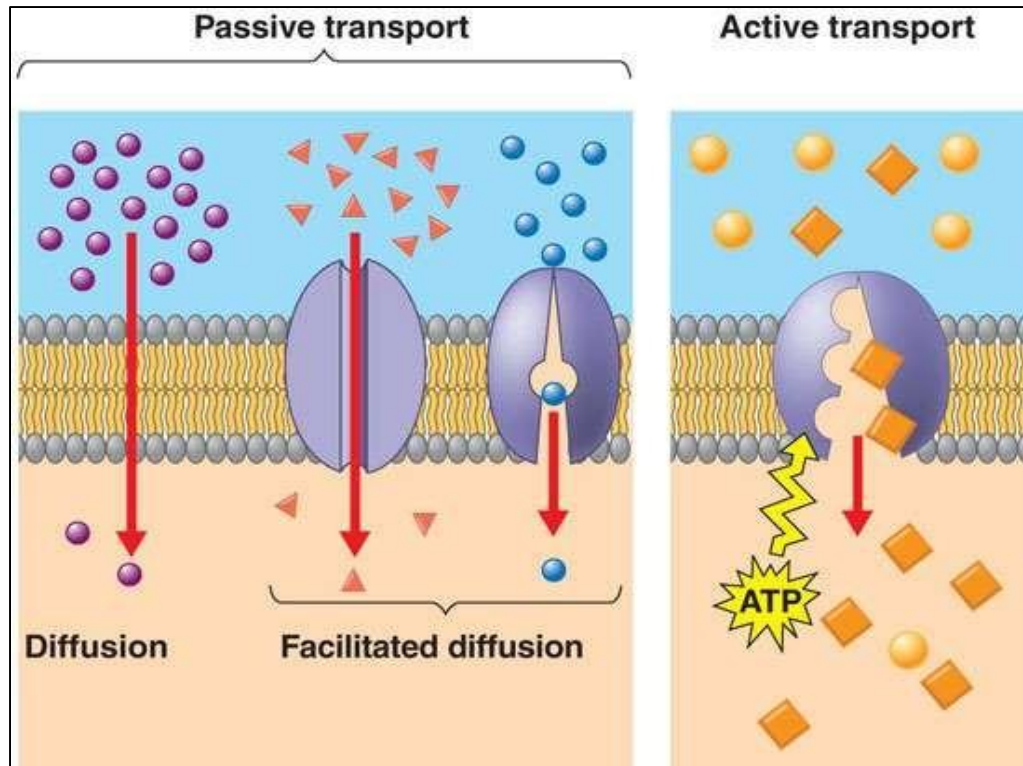
Q_d = activation energy

R = gas constant

T = absolute temperature

DIFFUSION ACROSS MEMBRANE

Diffusion across membrane:



There are **two ways** in which substances can enter or leave a cell:

1) Passive

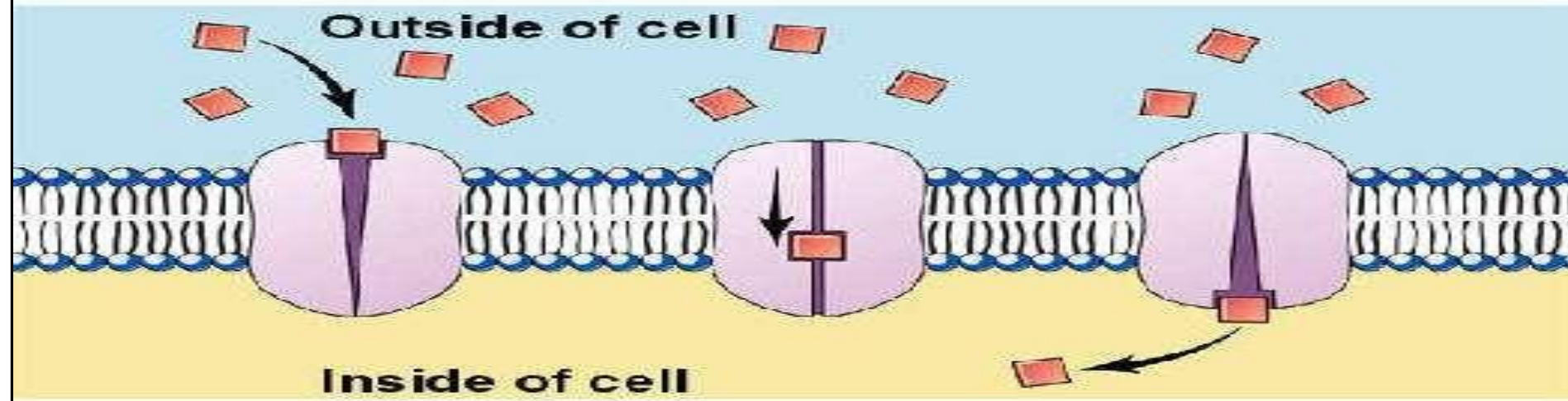
- a) Simple Diffusion
- b) Facilitated Diffusion
- c) Osmosis

2) Active

- a) Molecules
- b) Particles

DIFFUSION ACROSS MEMBRANE

Facilitated Diffusion



This is the movement of specific molecules down a concentration gradient, passing through the membrane via a specific carrier protein



Selection is by size; shape; charge.



Common molecules entering/leaving cells include glucose & amino-acids.

DIFFUSION ACROSS MEMBRANE



It is passive and requires no energy from the cell.

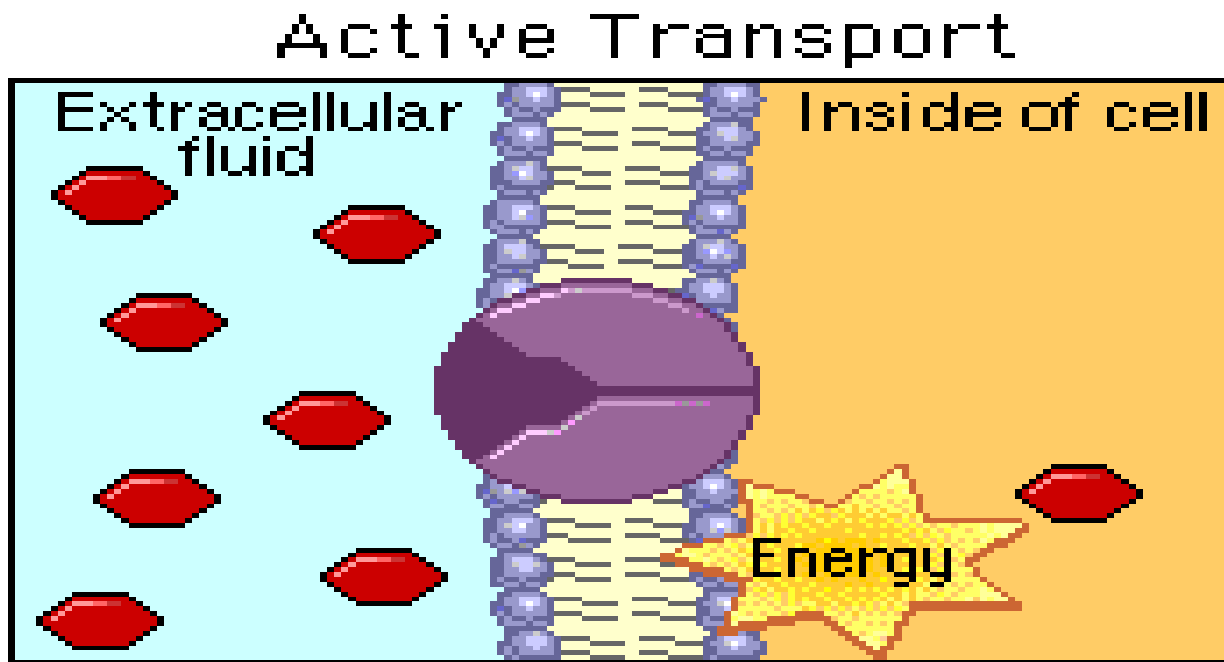


If the molecule is changed on entering the cell (glucose + ATP → glucose phosphate + ADP), then the concentration gradient of glucose will be kept high.

DIFFUSION ACROSS MEMBRANE

Active Transport :

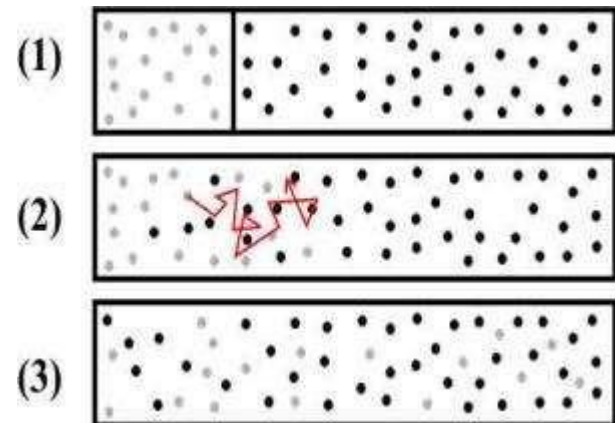
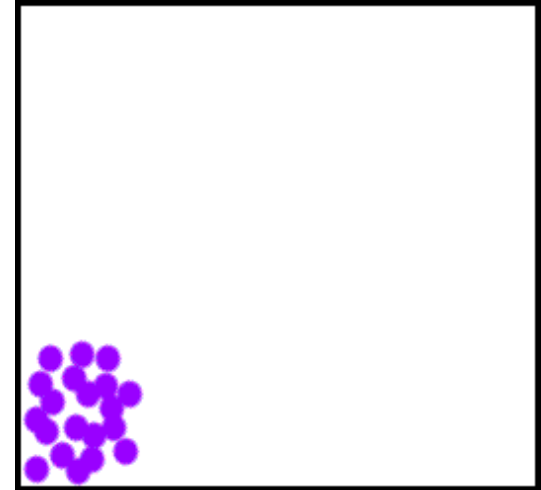
- Active transport is the **energy-demanding transfer** of a substance across a cell membrane **against its concentration gradient, i.e., from lower concentration to higher concentration.**
- Special proteins within the cell membrane act as specific protein 'carriers'. **The energy for active transport comes from ATP.**



DIFFUSION ACROSS MEMBRANE

Passive Transport:

- Passive diffusion is the process by which molecules spontaneously diffuse from a region of higher concentration to a region of lower concentration.
- Lipid-soluble drugs penetrate the lipid cell membrane and can pass the cell membrane by passive diffusion.
- Also, large molecules, such as proteins and protein-bound drugs, cannot diffuse through the cell membrane.



METHODS & PROCEDURES

Two types:

A) horizontal transport cell:

- a. wurester cell
- b. Viles chein permeation cell

B) vertical transport cell:

- a) Aquair and weiner diffusion cell
- b) biber and rhodes cell
- c) franz diffusion cell

APPLICATIONS

- 1) The release of drug from dosage form is diffusion controlled, such dosage form available in market as sustained & controlled release product.
- 2) The molecular weight of polymers can be estimated from diffusion process.
- 3) The transport of drugs from GI tract, skin, etc., can be understood & predicted from the principles of diffusion.
- 4) The diffusion of drugs into tissues & their excretion through kidneys can be anticipated through diffusion studies.
- 5) The process such as dialysis, microfiltration, ultrafiltration, haemodialysis, osmosis, etc., use the principles of diffusion.